CIS Apache Cassandra 4.0 Benchmark

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Overview

All CIS Benchmarks focus on technical configuration settings used to maintain and/or increase the security of the addressed technology, and they should be used in **conjunction** with other essential cyber hygiene tasks like:

- Monitoring the base operating system for vulnerabilities and quickly updating with the latest security patches
- Monitoring applications and libraries for vulnerabilities and quickly updating with the latest security patches

In the end, the CIS Benchmarks are designed as a key **component** of a comprehensive cybersecurity program.

This document, CIS Apache Cassandra Benchmark, provides prescriptive guidance for establishing a secure configuration posture for Apache Cassandra version 4.0. This guide was tested against Apache Cassandra running on CentOS Linux 7, but applies to other Linux distributions as well. To obtain the latest version of this guide, please visit http://benchmarks.cisecurity.org. If you have questions, comments, or have identified ways to improve this guide, please write us at feedback@cisecurity.org.

Intended Audience

This document is intended for system and application administrators, security specialists, auditors, help desk, and platform deployment personnel who plan to develop, deploy, assess, or secure solutions that incorporate Apache Cassandra.

Consensus Guidance

This CIS Benchmark was created using a consensus review process comprised of a global community of subject matter experts. The process combines real world experience with data-based information to create technology specific guidance to assist users to secure their environments. Consensus participants provide perspective from a diverse set of backgrounds including consulting, software development, audit and compliance, security research, operations, government, and legal.

Each CIS Benchmark undergoes two phases of consensus review. The first phase occurs during initial Benchmark development. During this phase, subject matter experts convene to discuss, create, and test working drafts of the Benchmark. This discussion occurs until consensus has been reached on Benchmark recommendations. The second phase begins after the Benchmark has been published. During this phase, all feedback provided by the Internet community is reviewed by the consensus team for incorporation in the Benchmark. If you are interested in participating in the consensus process, please visit https://workbench.cisecurity.org/.

Typographical Conventions

The following typographical conventions are used throughout this guide:

Convention	Meaning
`Stylized Monospace font`	Used for blocks of code, command, and script examples. Text should be interpreted exactly as presented.
`Monospace font`	Used for inline code, commands, or examples. Text should be interpreted exactly as presented.
<italic font in brackets>	Italic texts set in angle brackets denote a variable requiring substitution for a real value.
Italic font	Used to denote the title of a book, article, or other publication.
Note	Additional information or caveats

Recommendation Definitions

The following defines the various components included in a CIS recommendation as applicable. If any of the components are not applicable it will be noted or the component will not be included in the recommendation.

Title

Concise description for the recommendation's intended configuration.

Assessment Status

An assessment status is included for every recommendation. The assessment status indicates whether the given recommendation can be automated or requires manual steps to implement. Both statuses are equally important and are determined and supported as defined below:

Automated

Represents recommendations for which assessment of a technical control can be fully automated and validated to a pass/fail state. Recommendations will include the necessary information to implement automation.

Manual

Represents recommendations for which assessment of a technical control cannot be fully automated and requires all or some manual steps to validate that the configured state is set as expected. The expected state can vary depending on the environment.

Profile

A collection of recommendations for securing a technology or a supporting platform. Most benchmarks include at least a Level 1 and Level 2 Profile. Level 2 extends Level 1 recommendations and is not a standalone profile. The Profile Definitions section in the benchmark provides the definitions as they pertain to the recommendations included for the technology.

Description

Detailed information pertaining to the setting with which the recommendation is concerned. In some cases, the description will include the recommended value.

Rationale Statement

Detailed reasoning for the recommendation to provide the user a clear and concise understanding on the importance of the recommendation.

Impact Statement

Any security, functionality, or operational consequences that can result from following the recommendation.

Audit Procedure

Systematic instructions for determining if the target system complies with the recommendation

Remediation Procedure

Systematic instructions for applying recommendations to the target system to bring it into compliance according to the recommendation.

Default Value

Default value for the given setting in this recommendation, if known. If not known, either not configured or not defined will be applied.

References

Additional documentation relative to the recommendation.

CIS Critical Security Controls® (CIS Controls®)

The mapping between a recommendation and the CIS Controls is organized by CIS Controls version, Safeguard, and Implementation Group (IG). The Benchmark in its entirety addresses the CIS Controls safeguards of (v7) “5.1 - Establish Secure Configurations” and (v8) '4.1 - Establish and Maintain a Secure Configuration Process” so individual recommendations will not be mapped to these safeguards.

Additional Information

Supplementary information that does not correspond to any other field but may be useful to the user.

Profile Definitions

The following configuration profiles are defined by this Benchmark:

- **Level 1 - Cassandra**

  Items in this profile apply to Apache Cassandra and intend to:

  - be practical and prudent;
  - provide a clear security benefit; and
  - not inhibit the utility of the technology beyond acceptable means.

  Note: The intent of this profile is to include checks that can be assessed by remotely connecting to Cassandra. Therefore, file system-related checks are not contained in this profile.

- **Level 2 - Cassandra**

  This profile extends the “Level 1 - Cassandra” profile. Items in this profile apply to Apache Cassandra and exhibit one or more of the following characteristics:

  - are intended for environments or use cases where security is paramount
  - acts as defense in depth measure
  - may negatively inhibit the utility or performance of the technology.

  Note: The intent of this profile is to include checks that can be assessed by remotely connecting to Cassandra. Therefore, file system-related checks are not contained in this profile.

- **Level 1 - Cassandra on Linux**

  This profile extends the “Level 1 - Cassandra” profile. Items in this profile apply to Apache Cassandra running on Linux and intend to:

  - be practical and prudent;
  - provide a clear security benefit; and
  - not inhibit the utility of the technology beyond acceptable means.

- **Level 2 - Cassandra on Linux**

  This profile extends the “Level 1 - Cassandra on Linux” profile. Items in this profile apply to Apache Cassandra running on Linux and exhibit one or more of the following characteristics:

- are intended for environments or use cases where security is paramount
- acts as defense in depth measure
- may negatively inhibit the utility or performance of the technology.

Acknowledgements

This Benchmark exemplifies the great things a community of users, vendors, and subject matter experts can accomplish through consensus collaboration. The CIS community thanks the entire consensus team with special recognition to the following individuals who contributed greatly to the creation of this guide:

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Recommendations

1 Installation and Updates

This section contains recommendations related to installing and patching Cassandra.

1.1 Ensure a separate user and group exist for Cassandra (Manual)

Profile Applicability:

- Level 1 - Cassandra on Linux

Description:

Create separate userid and group for Cassandra.

Rationale:

All processes need to run as a user with least privilege. This mitigates the potential impact of malware to the system.

Audit:

Logon to the server where Cassandra is installed.
To confirm existence of the group, execute the following command:

```
$ getent group | grep cassandra
```

To confirm existence of the user, execute the following command:

```
$ getent passwd | grep cassandra
```

If either the group or user do not exist, or if the user is not a member of the group, this is a finding.

Remediation:

Create a group for cassandra(if it does not already exist)

```
sudo groupadd cassandra
```

Create a user which is only used for running Cassandra and its related processes.

```
sudo useradd -m -d /home/cassandra -s /bin/bash -g cassandra -u
<USERID_NUMBER> cassandra
```

Replacing *<USERID_NUMBER>* with a number not already used on the server

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 Configure Data Access Control Lists Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.	●	●	●

Controls Version	Control	IG 1	IG 2	IG 3
v8	5.4 Restrict Administrator Privileges to Dedicated Administrator Accounts Restrict administrator privileges to dedicated administrator accounts on enterprise assets. Conduct general computing activities, such as internet browsing, email, and productivity suite use, from the user's primary, non-privileged account.	●	●	●
v7	4.3 Ensure the Use of Dedicated Administrative Accounts Ensure that all users with administrative account access use a dedicated or secondary account for elevated activities. This account should only be used for administrative activities and not internet browsing, email, or similar activities.	●	●	●
v7	14.6 Protect Information through Access Control Lists Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.	●	●	●

1.2 Ensure the latest version of Java is installed (Automated)

Profile Applicability:

- Level 1 - Cassandra on Linux

Description:

A prerequisite to installing Cassandra is the installation of Java. The version of Java installed should be the most recent that is compatible with the organization's operational needs.

Rationale:

Using the most recent Java SDK version can help limit the possibilities for vulnerabilities in the software, the installation version applied during setup should be established according to the needs of the organization. Ensure you are using a release that is covered by a level of support which includes regular updates to address vulnerabilities.

Audit:

To verify that you have the correct version of java installed:

```
# java -version
java version "1.8.0_172"
Java(TM) SE Runtime Environment (build 1.8.0_172-b11)
```

If an old/unsupported version of Java is installed this is a finding.
Apache Cassandra expects a version of 1.8.
NOTE: Experimental support for Java 11 was added in Cassandra 4.0 (CASSANDRA-9608). Running Cassandra on Java 11 is experimental. Do so at your own risk.

Remediation:

1. Uninstall the old/unsupported version of Java, if present.
2. Download the latest compatible release of the Java JDK, or OpenJDK.
3. Follow the provided installation instructions to complete the install.

References:

1. http://www.oracle.com/technetwork/java/javase/downloads/index-jsp-138363.html#javasejdk
2. http://openjdk.java.net/
3. http://openjdk.java.net/install/index.html
4. http://cassandra.apache.org/doc/latest/getting_started/installing.html#prerequisites
5. https://www.java.com/en/download/help/index_installing.xml?os=All+Platforms&j=8&n=20

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	16.5 Use Up-to-Date and Trusted Third-Party Software Components Use up-to-date and trusted third-party software components. When possible, choose established and proven frameworks and libraries that provide adequate security. Acquire these components from trusted sources or evaluate the software for vulnerabilities before use.		●	●
v7	18.4 Only Use Up-to-date And Trusted Third-Party Components Only use up-to-date and trusted third-party components for the software developed by the organization.		●	●

1.3 Ensure the latest version of Python is installed (Automated)

Profile Applicability:

- Level 1 - Cassandra on Linux

Description:

A prerequisite to installing Cassandra is the installation of Python. The version of Python installed should be the most recent that is compatible with the organizations' operational needs.

Rationale:

Using the most recent Python can help limit the possibilities for vulnerabilities in the software, the installation version applied during setup should be established according to the needs of the organization. Ensure you are using a release that is covered by a level of support which includes regular updates to address vulnerabilities.

Audit:

To verify that you have the correct version of python installed:

```
# python -V
or
# python --version
```

If an old/unsupported version of Python is installed this is a finding.
For using cqlsh, the latest version of Python 3.6+ or Python 2.7 (support deprecated) is required.

Remediation:

1. Uninstall the old/unsupported version of Python, if present.
2. Download the latest compatible release of the Python: https://www.python.org/downloads/
3. Follow the provided installation instructions to complete the install.

References:

1. https://www.python.org/downloads/
2. http://cassandra.apache.org/doc/latest/getting_started/installing.html#prerequisites

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	16.5 Use Up-to-Date and Trusted Third-Party Software Components Use up-to-date and trusted third-party software components. When possible, choose established and proven frameworks and libraries that provide adequate security. Acquire these components from trusted sources or evaluate the software for vulnerabilities before use.		●	●
v7	18.4 Only Use Up-to-date And Trusted Third-Party Components Only use up-to-date and trusted third-party components for the software developed by the organization.		●	●

1.4 Ensure latest version of Cassandra is installed (Automated)

Profile Applicability:

- Level 1 - Cassandra on Linux

Description:

The Cassandra installation version, along with the patches, should be the most recent that is compatible with organization's operational needs. When obtaining and installing software packages (typically via apt-get or you can compile the source code), it's imperative that packages (or the source code, tarball) are sourced only from valid and authorized repositories.

For Cassandra, a short list of valid repositories may include:

- The official apache cassandra website: http://cassandra.apache.org/
- DataStax Enterprise: https://www.datastax.com/

Rationale:

Using the most recent version of Cassandra can help limit the possibilities for vulnerabilities in the software, the installation version applied during setup should be established according to the needs of the organization. Ensure you are using a release that is covered by a level of support which includes regular updates to address vulnerabilities.

Audit:

To verify the version of Cassandra you have installed:

```
cassandra -v

4.0.3 (a/o 2022-03-29)
```

Released on 2022-02-17
Maintained until 4.3.0 release (May-July 2024)
If an old/unsupported version of Cassandra is installed this is a finding.

Remediation:

Upgrade to the latest version of the Cassandra software:
For each node in the cluster:

1. Using the nodetool drain command to push all memtables data to SSTables.
2. Stop Cassandra services.
3. Backup the data set and all of your Cassandra configuration files.
4. Download/Update Java if needed.
5. Download/Update Python if needed.

6. Download the binaries for the latest Cassandra revision from the Cassandra Download Page.
7. Install new version of Cassandra.
8. Configure new version of Cassandra, taking into account all of your previous settings in your config files(`cassandra.yml`, `cassandrea-env.sh`, etc).
9. Start Cassandra services.
10. Check logs for warnings, errors.
11. Using the nodetool to upgrade your SSTables.
12. Using the nodetool command to check status of cluster.

References:

1. http://cassandra.apache.org/doc/latest/getting_started/installing.html#prerequisites

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	16.5 Use Up-to-Date and Trusted Third-Party Software Components Use up-to-date and trusted third-party software components. When possible, choose established and proven frameworks and libraries that provide adequate security. Acquire these components from trusted sources or evaluate the software for vulnerabilities before use.		●	●
v7	18.4 Only Use Up-to-date And Trusted Third-Party Components Only use up-to-date and trusted third-party components for the software developed by the organization.		●	●

1.5 Ensure the Cassandra service is run as a non-root user (Automated)

Profile Applicability:

- Level 1 - Cassandra on Linux

Description:

Though Cassandra database may be run as root, it should run as another non-root user.

Rationale:

One of the best ways to reduce your exposure to attack is to create a unique, unprivileged user and group for the server application. A best practice is to follow is ensuring processes run with a user with least privilege.

Audit:

Logon to the server where Cassandra is running and run the following command

```
ps -aef | grep cassandra | grep java | cut -d' ' -f1
```

This will show who is running the Cassandra binary.
If the user is root or has excessive privileges then this is a finding.

Remediation:

Create a group for cassandra (if it does not already exist)

```
sudo groupadd cassandra
```

Create a user which is only used for running Cassandra and its related processes.

Replacing *<DIRECTORY_WHERE_CASSANDRA_INSTALLED>* with the full path of where Cassandra binaries are installed.
Replacing *<USERID_NUMBER>* with a number not already used on the server

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 Configure Data Access Control Lists Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.	●	●	●

Controls Version	Control	IG 1	IG 2	IG 3
v8	5.4 Restrict Administrator Privileges to Dedicated Administrator Accounts Restrict administrator privileges to dedicated administrator accounts on enterprise assets. Conduct general computing activities, such as internet browsing, email, and productivity suite use, from the user's primary, non-privileged account.	●	●	●
v7	4.3 Ensure the Use of Dedicated Administrative Accounts Ensure that all users with administrative account access use a dedicated or secondary account for elevated activities. This account should only be used for administrative activities and not internet browsing, email, or similar activities.	●	●	●
v7	14.6 Protect Information through Access Control Lists Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.	●	●	●

1.6 Ensure clocks are synchronized on all nodes (Manual)

Profile Applicability:

- Level 1 - Cassandra on Linux

Description:

Enabling Network Time Protocol (NTP), or some equivalent way, to keep clocks on all nodes in sync is critical.

Rationale:

Cassandra decides which data is most current between all of the nodes in the cluster based on timestamps. It is paramount to ensure all clocks are in-sync, otherwise the most current data may not be returned or worse, marked for deletion.

Audit:

Depending on the Linux installation this may be checked by executing the following command on each node:

```
ps -aef | grep ntp

OR

ps -aef | grep chronyd

OR

timedatectl status | grep NTP
```

If NTP is not configured or clocks are out-of-sync then this is a finding.

Remediation:

Install and start the time protocol on every node in the Cassandra cluster.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.4 Standardize Time Synchronization Standardize time synchronization. Configure at least two synchronized time sources across enterprise assets, where supported.		●	●
v7	6.1 Utilize Three Synchronized Time Sources Use at least three synchronized time sources from which all servers and network devices retrieve time information on a regular basis so that timestamps in logs are consistent.		●	●

2 Authentication and Authorization

This section contains recommendations related to Cassandra's authentication and authorization mechanisms.

2.1 Ensure that authentication is enabled for Cassandra databases (Automated)

Profile Applicability:

- Level 1 - Cassandra on Linux

Description:

Authentication is pluggable in Cassandra and is configured using the `authenticator` setting in `cassandra.yaml`. Cassandra ships with two options included in the default distribution, `AllowAllAuthenticator` and `PasswordAuthenticator`. The default, `AllowAllAuthenticator`, performs no authentication checks and therefore requires no credentials. It is used to disable authentication completely. The second option, `PasswordAuthenticator`, stores encrypted credentials in a system table. This can be used to enable simple username/password authentication.

Rationale:

Authentication is a necessary condition of Cassandra's permissions subsystem, so if authentication is disabled then so are permissions. Failure to authenticate clients, users, and/or servers can allow unauthorized access to the Cassandra database and can prevent tracing actions back to their sources. The authentication mechanism should be implemented before anyone accesses the Cassandra server.

Audit:

Run the following command to verify whether authentication is enabled (authenticator values set to `PasswordAuthenticator`) on the Cassandra server.
The Cassandra configuration files can be found in the conf directory of tarballs. For packages, the configuration files will be located in `/etc/cassandra`.

```
cat cassandra.yaml | grep -in "authenticator:"
```

If `authenticator` is set to `AllowAllAuthenticator`, then this is a finding.

Remediation:

To enable the authentication mechanism:

1. Stop the Cassandra database.
2. Modify `cassandra.yaml` file to modify/add entry for authenticator: set it to `PasswordAuthenticator`
3. Start the Cassandra database.

Default Value:

`authenticator: AllowAllAuthenticator`

References:

1. http://cassandra.apache.org/doc/latest/getting_started/configuring.html
2. http://cassandra.apache.org/doc/latest/operating/security.html

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	16.11 Leverage Vetted Modules or Services for Application Security Components Leverage vetted modules or services for application security components, such as identity management, encryption, and auditing and logging. Using platform features in critical security functions will reduce developers' workload and minimize the likelihood of design or implementation errors. Modern operating systems provide effective mechanisms for identification, authentication, and authorization and make those mechanisms available to applications. Use only standardized, currently accepted, and extensively reviewed encryption algorithms. Operating systems also provide mechanisms to create and maintain secure audit logs.		●	●
v7	14.7 Enforce Access Control to Data through Automated Tools Use an automated tool, such as host-based Data Loss Prevention, to enforce access controls to data even when data is copied off a system.			●

2.2 Ensure that authorization is enabled for Cassandra databases (Automated)

Profile Applicability:

- Level 1 - Cassandra on Linux

Description:

Authorization is pluggable in Cassandra and is configured using the `authorizer` setting in `cassandra.yaml`. Cassandra ships with two options included in the default distribution, `AllowAllAuthenticator` and `CassandraAuthorizer`. The default, `AllowAllAuthenticator` performs no checking which grants all permissions to all roles. The second option, `CassandraAuthorizer`, implements full permissions management functionality and stores its data in Cassandra system tables.

Rationale:

Authorizing roles is an important step towards ensuring only authorized access to the Cassandra database tables is permitted. It also provides the requisite means of implementing least privilege best practices. The authorization mechanism should be implemented before anyone accesses the Cassandra database.

Audit:

Run the following command to verify whether authorization is enabled (authorization values set to `CassandraAuthorizer`) on the Cassandra server.
The Cassandra configuration files can be found in the conf directory of tarballs. For packages, the configuration files will be located in /etc/cassandra.

```
cat cassandra.yaml | grep -in "authorizer:"
```

If `authorizer` is set to `AllowAllAuthorizer`, then this is a finding.

Remediation:

To enable the authorization mechanism:

1. Stop the Cassandra database.
2. Modify cassandra.yaml file to modify/add entry for authorization: set it to CassandraAuthorizer
3. Start the Cassandra database.

Default Value:

`authorizer: AllowAllAuthorizer`

References:

1. http://cassandra.apache.org/doc/latest/getting_started/configuring.html

2. http://cassandra.apache.org/doc/latest/operating/security.html

Additional Information:

The `authorizer` must be configured to `AllowAllAuthorizer` if `AllowAllAuthenticator` is the configured authenticator.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	16.11 Leverage Vetted Modules or Services for Application Security Components Leverage vetted modules or services for application security components, such as identity management, encryption, and auditing and logging. Using platform features in critical security functions will reduce developers' workload and minimize the likelihood of design or implementation errors. Modern operating systems provide effective mechanisms for identification, authentication, and authorization and make those mechanisms available to applications. Use only standardized, currently accepted, and extensively reviewed encryption algorithms. Operating systems also provide mechanisms to create and maintain secure audit logs.		●	●
v7	14.7 Enforce Access Control to Data through Automated Tools Use an automated tool, such as host-based Data Loss Prevention, to enforce access controls to data even when data is copied off a system.			●

3 Access Control / Password Policies

This section contains recommendations related to Cassandra's password policies.

3.1 Ensure the cassandra and superuser roles are separate (Automated)

Profile Applicability:

- Level 1 - Cassandra
- Level 1 - Cassandra on Linux

Description:

The default installation of Cassandra includes a superuser role named `cassandra`. This necessitates the creation of a separate role to be the superuser role.

Rationale:

Superuser permissions allow for the creation, deletion, and permission management of other users. Considering the cassandra role is well known it should not be a superuser or one which is used for any administrative tasks.

Impact:

If a separate superuser account is not created and tested for correct functionality prior to removing the superuser role from the `cassandra` account you will no longer be able to perform certain actions, including:

- Create a role with super user status.
- Perform `DROP` or `CREATE USER` queries.

Audit:

To verify the configuration, run the following query:

```
select role from system_auth.roles where is_superuser= True;
```

If you get an error 2200 [INVALID QUERY] due to where clause, add "ALLOW FILTERING to end of query so it looks like this:

```
select role from system_auth.roles where is_superuser= True ALLOW FILTERING;
```

Looking at the role, verify any show up with is_superuser = True and make sure it is not `cassandra` or any unapproved role. If any are found then, this is a finding.

Remediation:

To remediate a misconfiguration, perform the following steps:

1. Execute the following command:

```
create role '<NEW_ROLE_HERE>' with password='<NEW_PASSWORD_HERE>' and
login=TRUE and superuser=TRUE ;

grant all permissions on all keyspaces to <NEW_ROLE_HERE>;
```

Note: Replace *<NEW_ROLE_HERE>* with the desired role and *<NEW_PASSWORD_HERE>* with a password.

2. Verify the new role is working.
3. Remove the superuser role from the `cassandra` account by executing the following command:

```
UPDATE system_auth.roles SET is_superuser=null WHERE role='cassandra';
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	5.4 Restrict Administrator Privileges to Dedicated Administrator Accounts Restrict administrator privileges to dedicated administrator accounts on enterprise assets. Conduct general computing activities, such as internet browsing, email, and productivity suite use, from the user's primary, non-privileged account.	●	●	●
v7	4.3 Ensure the Use of Dedicated Administrative Accounts Ensure that all users with administrative account access use a dedicated or secondary account for elevated activities. This account should only be used for administrative activities and not internet browsing, email, or similar activities.	●	●	●

3.2 Ensure that the default password is changed for the cassandra role (Automated)

Profile Applicability:

- Level 1 - Cassandra
- Level 1 - Cassandra on Linux

Description:

The cassandra role has a default password which must be changed.

Rationale:

Failure to change the default password for the cassandra role may pose a risk to the database in the form of unauthorized access.

Audit:

Connect to Cassandra database to verify whether the cassandra role has default password.

```
cqlsh -u cassandra -p cassandra
```

If the connection is successful this is a finding.

Remediation:

Change the password for the casssandra role by issuing the following command:

```
cqlsh -u cassandra -p cassandra

alter role 'cassandra' with password '<NEWPASSWORD_HERE>';
```

Where *<NEWPASSWORD_HERE>* is replaced with the password of your choosing.

Default Value:

cassandra

References:

1. http://cassandra.apache.org/doc/latest/operating/security.html

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	5.2 Use Unique Passwords Use unique passwords for all enterprise assets. Best practice implementation includes, at a minimum, an 8-character password for accounts using MFA and a 14-character password for accounts not using MFA.	●	●	●
v7	4.4 Use Unique Passwords Where multi-factor authentication is not supported (such as local administrator, root, or service accounts), accounts will use passwords that are unique to that system.		●	●

3.3 Ensure there are no unnecessary roles or excessive privileges (Manual)

Profile Applicability:

- Level 1 - Cassandra
- Level 1 - Cassandra on Linux

Description:

Verify each role is require and has only the privileges needed to do its job.

Rationale:

Roles which are unneeded, have super user or other potentially excessive privileges may be an avenue for a hacker to gain access to or modify data in the database.

Audit:

As a superuser, retrieve all roles:

```
list roles;
```

Retrieve all permissions for all roles

```
select * from system_auth.role_permissions;
```

If there are any unnecessary roles or roles with excessive privileges this is a finding.

Remediation:

Remove any unnecessary roles and/or permissions in accordance with organizational needs.

References:

1. http://cassandra.apache.org/doc/latest/cql/security.html

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	6.8 Define and Maintain Role-Based Access Control Define and maintain role-based access control, through determining and documenting the access rights necessary for each role within the enterprise to successfully carry out its assigned duties. Perform access control reviews of enterprise assets to validate that all privileges are authorized, on a recurring schedule at a minimum annually, or more frequently.			●

Controls Version	Control	IG 1	IG 2	IG 3
v7	14.6 Protect Information through Access Control Lists Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.	●	●	●

3.4 Ensure that Cassandra is run using a non-privileged, dedicated service account (Automated)

Profile Applicability:

- Level 1 - Cassandra on Linux

Description:

As with any service installed on a host, it can be provided with its own user context. Providing a dedicated user to the service provides the ability to precisely constrain the service within the larger host context.

Rationale:

Utilizing a non-privileged account for Cassandra to execute as may reduce the impact of a Cassandra-born vulnerability. A restricted account will be unable to access resources unrelated to Cassandra, such as operating system configurations.

Audit:

Execute the following command at a terminal prompt to assess this recommendation:

```
ps -ef | egrep "^cassandra.*$"
```

If no lines are returned, then this is a finding.
NOTE: It is assumed that the Cassandra user is `cassandra`. Additionally, you may consider running `sudo -l` as the Cassandra user or to check the `sudoers` file.

Remediation:

Create a user which is only used for running Cassandra and directly related processes. This user must not have administrative rights to the system.

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.3 Configure Data Access Control Lists Configure data access control lists based on a user's need to know. Apply data access control lists, also known as access permissions, to local and remote file systems, databases, and applications.	●	●	●
v8	5.4 Restrict Administrator Privileges to Dedicated Administrator Accounts Restrict administrator privileges to dedicated administrator accounts on enterprise assets. Conduct general computing activities, such as internet browsing, email, and productivity suite use, from the user's primary, non-privileged account.	●	●	●

Controls Version	Control	IG 1	IG 2	IG 3
v7	4.3 Ensure the Use of Dedicated Administrative Accounts Ensure that all users with administrative account access use a dedicated or secondary account for elevated activities. This account should only be used for administrative activities and not internet browsing, email, or similar activities.	●	●	●
v7	14.6 Protect Information through Access Control Lists Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.	●	●	●

3.5 Ensure that Cassandra only listens for network connections on authorized interfaces (Manual)

Profile Applicability:

- Level 1 - Cassandra on Linux

Description:

When `listen_address` is blank and `listen_interface` is commented out, this will be set automatically by `InetAddress.getLocalHost()`. Presuming the node is configured correctly, e.g. hostname, name resolution, etc., this will configure the node to use the address associated with the hostname. The `listen_address` must not be set to `0.0.0.0`.

Rationale:

Setting the address or interface to bind to will tell other Cassandra nodes to which address or interface to connect. This must be changed from the default in order for multiple nodes to be able to communicate.

Audit:

Check the value of `listen_address` or `listen_interface` in the `cassandra.yaml`. If `listen_address` is set `0.0.0.0` or a non-authorized address or interface is specified, this is a finding.

Remediation:

Set the `listen_address` or `listen_interface`, not both, in the `cassandra.yaml` to an authorized address or interface.

Default Value:

`listen_address`: `localhost`

`listen_interface`: `eth0`, but is commented out by default.

References:

1. http://cassandra.apache.org/doc/3.11/configuration/cassandra_config_file.html#listen-address
2. http://cassandra.apache.org/doc/3.11/configuration/cassandra_config_file.html#listen-interface

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	4.4 Implement and Manage a Firewall on Servers Implement and manage a firewall on servers, where supported. Example implementations include a virtual firewall, operating system firewall, or a third-party firewall agent.	●	●	●
v7	9.2 Ensure Only Approved Ports, Protocols and Services Are Running Ensure that only network ports, protocols, and services listening on a system with validated business needs, are running on each system.		●	●

3.6 Ensure that Data Center Authorizations is activated (Manual)

Profile Applicability:

- Level 1 - Cassandra
- Level 1 - Cassandra on Linux

Description:

Authorization at Data Center level is pluggable in Cassandra and is configured using the `network_authorizer` setting in `cassandra.yaml`. Cassandra ships with `AllowAllNetworkAuthorizer` which allows any role to access any datacenter effectively disabling datacenter authorization; which is the current behavior.

It should be set to `CassandraNetworkAuthorizer` which allows the ability to store permissions which restrict role access to specific datacenters.

Rationale:

The `network_authorizer` parameter in the cassandra.yaml file allows an operator to restrict the access of a Cassandra role to specific datacenters. Keep in mind that for this to work correctly, the authenticator setting in `cassandra.yaml` file must be set to `PasswordAuthenticator`.

Audit:

Run the following command to verify whether network authorization is enabled (`network_authorizer` value set to `CassandraNetworkAuthorizer`) on the Cassandra server.
The Cassandra configuration files can be found in the conf directory of tarballs. For packages, the configuration files will be located in /etc/cassandra.

```
cat cassandra.yaml | grep -in "network_authorizer:"
```

Remediation:

1. Stop the Cassandra database on each node.
2. Modify the cassandra.yaml file to modify entry for network_authorizer: set it to CassandraNetworkAuthorizer
3. Start the Cassandra database.

Default Value:

`network_authorizer: AllowAllNetworkAuthorizer`

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	16.11 Leverage Vetted Modules or Services for Application Security Components Leverage vetted modules or services for application security components, such as identity management, encryption, and auditing and logging. Using platform features in critical security functions will reduce developers' workload and minimize the likelihood of design or implementation errors. Modern operating systems provide effective mechanisms for identification, authentication, and authorization and make those mechanisms available to applications. Use only standardized, currently accepted, and extensively reviewed encryption algorithms. Operating systems also provide mechanisms to create and maintain secure audit logs.		●	●
v7	14.7 Enforce Access Control to Data through Automated Tools Use an automated tool, such as host-based Data Loss Prevention, to enforce access controls to data even when data is copied off a system.			●

3.7 Review User-Defined Roles (Manual)

Profile Applicability:

- Level 1 - Cassandra
- Level 1 - Cassandra on Linux

Description:

The `MEMBER_OF` column found in the `system_auth.roles` table shows roles granted to roles.

Rationale:

The `MEMBER_OF` column shows whoever has roles granted to roles and depending on the role and the privileges grant to the role should be limited . Limiting the accounts that have the certain roles reduces the chances that an attacker can exploit these capabilities.

Audit:

Execute the following SQL statement to audit this setting:

```
select role, can_login, member_of from system_auth.roles;
```

Looking for `can_login` which tells you that role can log into cassandra and `member_of` is when roles are granted to roles.

Remediation:

Looking at those users from the query that have member_of that is NOT null, decide if that user truly needs that role, if not, for each user, issue the following SQL statement (replace *`<is_member>`* with the value of `member_of` returned by the query in the audit procedure)

```
revoke <is_member> from role;
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	6.8 Define and Maintain Role-Based Access Control Define and maintain role-based access control, through determining and documenting the access rights necessary for each role within the enterprise to successfully carry out its assigned duties. Perform access control reviews of enterprise assets to validate that all privileges are authorized, on a recurring schedule at a minimum annually, or more frequently.			●

Controls Version	Control	IG 1	IG 2	IG 3
v7	14.6 Protect Information through Access Control Lists Protect all information stored on systems with file system, network share, claims, application, or database specific access control lists. These controls will enforce the principle that only authorized individuals should have access to the information based on their need to access the information as a part of their responsibilities.	●	●	●

3.8 Review Superuser/Admin Roles (Manual)

Profile Applicability:

- Level 1 - Cassandra
- Level 1 - Cassandra on Linux

Description:

The `IS_SUPERUSER` privilege found in the `system_auth.roles` table governs who can control the entire Cassandra database and all of its data contained within.

Rationale:

The `IS_SUPERUSER` privilege allows whoever has it to do anything to the data and full administrator rights to the database, including changing passwords, creating, dropping roles. Limiting the accounts that have the `IS_SUPERUSER` role reduces the chances that an attacker can exploit these capabilities.

Audit:

Execute the following SQL statement to audit this setting:

```
select role, is_superuser from system_auth.roles;
```

Looking for `is_superuser = True`

Remediation:

Perform the following steps to remediate this setting:
Looking at those users from the query that have `is_superuser = True`, decide if that user truly needs that role, if not, for each user, issue the following SQL statement (replace *`<role>`* with the role name from the query):

```
alter role <role> with superuser=false;
```

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	5.4 Restrict Administrator Privileges to Dedicated Administrator Accounts Restrict administrator privileges to dedicated administrator accounts on enterprise assets. Conduct general computing activities, such as internet browsing, email, and productivity suite use, from the user's primary, non-privileged account.	●	●	●

Controls Version	Control	IG 1	IG 2	IG 3
v7	4.3 Ensure the Use of Dedicated Administrative Accounts Ensure that all users with administrative account access use a dedicated or secondary account for elevated activities. This account should only be used for administrative activities and not internet browsing, email, or similar activities.	●	●	●

4 Auditing and Logging

This section contains recommendations related to Cassandra's audit and logging mechanisms.

4.1 Ensure that logging is enabled. (Automated)

Profile Applicability:

- Level 1 - Cassandra
- Level 1 - Cassandra on Linux

Description:

Apache Cassandra uses Logback for logging functionality. While this can be set using `nodetool setlogginglevel` changes made using this method will be reverted to the level specified in the `logback.xml` file the next time the process restarts.

The configurable logging levels are:

- `OFF`
- `TRACE`
- `DEBUG`
- `INFO` (Default)
- `WARN`
- `ERROR`

Rationale:

If logging is not enabled, issues may go undiscovered, and compromises and other incidents may occur without being quickly detected. It may also not be possible to provide evidence of compliance with security laws, regulations, and other requirements.

Audit:

Execute the following command to confirm the setting is correct:

```
$ nodetool getlogginglevels
Logger Name                                          Log Level
ROOT                                                      INFO
org.cisecurity.workbench                                  WARN
```

If set to `OFF` then this is a finding.

Remediation:

To remediate this setting:

1. Edit the `logback-test.xml` if present; otherwise, edit the `logback.xml`

```
<configuration scan="true">

    <appender name="STDOUT"
class="ch.qos.logback.core.ConsoleAppender">
        <filter class="ch.qos.logback.classic.filter.ThresholdFilter">
            <level>INFO</level>
        </filter>
        <encoder>
            <pattern>%-5level [%thread] %date{ISO8601} %F:%L -
%msg%n</pattern>
        </encoder>
    </appender>

    <root level="INFO">
        <appender-ref ref="STDOUT" />
    </root>

    <logger name="org.cisecurity.workbench" level="WARN"/>
</configuration>
```

2. Restart the Apache Cassandra

Default Value:

INFO

References:

1. http://cassandra.apache.org/doc/latest/troubleshooting/reading_logs.html?highlight=logging
2. https://logback.qos.ch/manual/configuration.html

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.5 Collect Detailed Audit Logs Configure detailed audit logging for enterprise assets containing sensitive data. Include event source, date, username, timestamp, source addresses, destination addresses, and other useful elements that could assist in a forensic investigation.		●	●
v7	6.3 Enable Detailed Logging Enable system logging to include detailed information such as an event source, date, user, timestamp, source addresses, destination addresses, and other useful elements.		●	●

4.2 Ensure that auditing is enabled (Manual)

Profile Applicability:

- Level 1 - Cassandra on Linux

Description:

Audit logging in Cassandra logs every incoming CQL command request, Authentication (successful as well as unsuccessful login) to C* node. Currently, there are two implementations provided, the custom logger can be implemented and injected with the class name as a parameter in cassandra.yaml.

Rationale:

Unauthorized attempts to create, drop or alter users or data should be a concern.

Audit:

Allows logging to filesystem log files using logback or to a Cassandra table. When you turn on audit logging, the default is to write to logback filesystem log files. You can verify auditing is turned on:

```
cat cassandra.yaml | grep "audit_logging_options"
```

If failure is enabled: `true` means success
Anything else is a finding.

Remediation:

Open the `cassandra.yaml` file in a text editor
In the `audit_logging_options` section, set `enabled` to `true`.

```
# Audit logging options
audit_logging_options:
    enabled: true
```

References:

1. https://docs.datastax.com/en/datastax_enterprise/4.8/datastax_enterprise/sec/secAudit.html#secAudit

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	8.2 Collect Audit Logs Collect audit logs. Ensure that logging, per the enterprise's audit log management process, has been enabled across enterprise assets.	●	●	●

Controls Version	Control	IG 1	IG 2	IG 3
v7	6.2 Activate audit logging Ensure that local logging has been enabled on all systems and networking devices.	●	●	●

5 Encryption

These recommendations pertain to encryption-related aspects of Cassandra.

5.1 Inter-node Encryption (Automated)

Profile Applicability:

- Level 1 - Cassandra on Linux

Description:

Cassandra offers the option to encrypt data in transit between nodes on the cluster. By default, inter-node encryption is turned off.

Rationale:

Data being transferred on the wire should be encrypted to avoid network snooping, whether legitimate or not.

Audit:

Run the following command to verify whether inter-node encryption is enabled.

```
cat cassandra.yaml | grep -in "internode_encryption:"
```

Acceptable values are `all`, `dc` or `rack`. If the `internode_encryption` is set to `none`, this is a finding.
Note: The Cassandra configuration files can be found in the conf directory of tarballs. For packages, the configuration files will be located in `/etc/cassandra`.

Remediation:

The inter-node encryption should be implemented before anyone accesses the Cassandra server.
To enable the inter-node encryption mechanism:

1. Stop the Cassandra database.
2. If not done so already, build out your keystore and truststore.
3. Modify `cassandra.yaml` file to modify/add entry for `internode_encryption:` set it to `all`
4. Start the Cassandra database.

Default Value:

`internode_encryption: none`

References:

1. http://cassandra.apache.org/doc/latest/operating/security.html

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.10 Encrypt Sensitive Data in Transit Encrypt sensitive data in transit. Example implementations can include: Transport Layer Security (TLS) and Open Secure Shell (OpenSSH).		●	●
v7	14.4 Encrypt All Sensitive Information in Transit Encrypt all sensitive information in transit.		●	●

5.2 Client Encryption (Automated)

Profile Applicability:

- Level 1 - Cassandra on Linux

Description:

Cassandra offers the option to encrypt data in transit between the client and nodes on the cluster. By default client encryption is turned off.

Rationale:

Data in transit between the client and node on the cluster should be encrypted to avoid network snooping, whether legitimate or not.

Audit:

The Cassandra configuration files can be found in the conf directory of tarballs. For packages, the configuration files will be located in `/etc/cassandra`.
Open up the `cassandra.yaml` file, look for client_encryption_options section.
Look for `enabled:` and `optional:`

```
enabled: true

optional: false
```

If neither is true, then all client connections are unencrypted which makes this a finding.
If enabled is true and optional is false, then all client connections must be encrypted which makes this not a finding.
If enabled is false and optional is true, then enabled wins and all client connections are unencrypted which makes this a finding.
If both are set to true, then both unencrypted and encrypted connections are allowed on the same port which makes this not a finding.

Remediation:

The client encryption should be implemented before anyone accesses the Cassandra server.
To enable the client encryption mechanism:

1. Stop the Cassandra database.
2. If not done so already, build out your keystore and truststore.
3. Modify `cassandra.yaml` file to modify/add entries under `client_encryption_options:`

```
set enabled: true

set optional: false
```

This will force all connections to be encrypted between client and node on the cluster.

4. Start the Cassandra database.

Default Value:

```
enabled: false

optional: false
```

References:

1. http://cassandra.apache.org/doc/latest/operating/security.html

CIS Controls:

Controls Version	Control	IG 1	IG 2	IG 3
v8	3.10 Encrypt Sensitive Data in Transit Encrypt sensitive data in transit. Example implementations can include: Transport Layer Security (TLS) and Open Secure Shell (OpenSSH).		●	●
v7	14.4 Encrypt All Sensitive Information in Transit Encrypt all sensitive information in transit.		●	●

Appendix: Summary Table

CIS Benchmark Recommendation		Set Correctly	
		Yes	No
1	**Installation and Updates**		
1.1	Ensure a separate user and group exist for Cassandra (Manual)	☐	☐
1.2	Ensure the latest version of Java is installed (Automated)	☐	☐
1.3	Ensure the latest version of Python is installed (Automated)	☐	☐
1.4	Ensure latest version of Cassandra is installed (Automated)	☐	☐
1.5	Ensure the Cassandra service is run as a non-root user (Automated)	☐	☐
1.6	Ensure clocks are synchronized on all nodes (Manual)	☐	☐
2	**Authentication and Authorization**		
2.1	Ensure that authentication is enabled for Cassandra databases (Automated)	☐	☐
2.2	Ensure that authorization is enabled for Cassandra databases (Automated)	☐	☐
3	**Access Control / Password Policies**		
3.1	Ensure the cassandra and superuser roles are separate (Automated)	☐	☐
3.2	Ensure that the default password is changed for the cassandra role (Automated)	☐	☐
3.3	Ensure there are no unnecessary roles or excessive privileges (Manual)	☐	☐
3.4	Ensure that Cassandra is run using a non-privileged, dedicated service account (Automated)	☐	☐

CIS Benchmark Recommendation		Set Correctly	
		Yes	No
3.5	Ensure that Cassandra only listens for network connections on authorized interfaces (Manual)	☐	☐
3.6	Ensure that Data Center Authorizations is activated (Manual)	☐	☐
3.7	Review User-Defined Roles (Manual)	☐	☐
3.8	Review Superuser/Admin Roles (Manual)	☐	☐
4	**Auditing and Logging**		
4.1	Ensure that logging is enabled. (Automated)	☐	☐
4.2	Ensure that auditing is enabled (Manual)	☐	☐
5	**Encryption**		
5.1	Inter-node Encryption (Automated)	☐	☐
5.2	Client Encryption (Automated)	☐	☐

Appendix: CIS Controls v7 IG 1 Mapped Recommendations

Recommendation		Set Correctly	
		Yes	No
1.1	Ensure a separate user and group exist for Cassandra	☐	☐
1.5	Ensure the Cassandra service is run as a non-root user	☐	☐
3.1	Ensure the cassandra and superuser roles are separate	☐	☐
3.3	Ensure there are no unnecessary roles or excessive privileges	☐	☐
3.4	Ensure that Cassandra is run using a non-privileged, dedicated service account	☐	☐
3.7	Review User-Defined Roles	☐	☐
3.8	Review Superuser/Admin Roles	☐	☐
4.2	Ensure that auditing is enabled	☐	☐

Appendix: CIS Controls v7 IG 2 Mapped Recommendations

Recommendation		Set Correctly	
		Yes	No
1.1	Ensure a separate user and group exist for Cassandra	☐	☐
1.2	Ensure the latest version of Java is installed	☐	☐
1.3	Ensure the latest version of Python is installed	☐	☐
1.4	Ensure latest version of Cassandra is installed	☐	☐
1.5	Ensure the Cassandra service is run as a non-root user	☐	☐
1.6	Ensure clocks are synchronized on all nodes	☐	☐
3.1	Ensure the cassandra and superuser roles are separate	☐	☐
3.2	Ensure that the default password is changed for the cassandra role	☐	☐
3.3	Ensure there are no unnecessary roles or excessive privileges	☐	☐
3.4	Ensure that Cassandra is run using a non-privileged, dedicated service account	☐	☐
3.5	Ensure that Cassandra only listens for network connections on authorized interfaces	☐	☐
3.7	Review User-Defined Roles	☐	☐
3.8	Review Superuser/Admin Roles	☐	☐
4.1	Ensure that logging is enabled.	☐	☐
4.2	Ensure that auditing is enabled	☐	☐
5.1	Inter-node Encryption	☐	☐
5.2	Client Encryption	☐	☐

Appendix: CIS Controls v7 IG 3 Mapped Recommendations

Recommendation		Set Correctly	
		Yes	No
1.1	Ensure a separate user and group exist for Cassandra	☐	☐
1.2	Ensure the latest version of Java is installed	☐	☐
1.3	Ensure the latest version of Python is installed	☐	☐
1.4	Ensure latest version of Cassandra is installed	☐	☐
1.5	Ensure the Cassandra service is run as a non-root user	☐	☐
1.6	Ensure clocks are synchronized on all nodes	☐	☐
2.1	Ensure that authentication is enabled for Cassandra databases	☐	☐
2.2	Ensure that authorization is enabled for Cassandra databases	☐	☐
3.1	Ensure the cassandra and superuser roles are separate	☐	☐
3.2	Ensure that the default password is changed for the cassandra role	☐	☐
3.3	Ensure there are no unnecessary roles or excessive privileges	☐	☐
3.4	Ensure that Cassandra is run using a non-privileged, dedicated service account	☐	☐
3.5	Ensure that Cassandra only listens for network connections on authorized interfaces	☐	☐
3.6	Ensure that Data Center Authorizations is activated	☐	☐
3.7	Review User-Defined Roles	☐	☐
3.8	Review Superuser/Admin Roles	☐	☐
4.1	Ensure that logging is enabled.	☐	☐
4.2	Ensure that auditing is enabled	☐	☐
5.1	Inter-node Encryption	☐	☐
5.2	Client Encryption	☐	☐

Appendix: CIS Controls v7 Unmapped Recommendations

Recommendation		Set Correctly	
		Yes	No
	No unmapped recommendations to CIS Controls v7.0	☐	☐

Appendix: CIS Controls v8 IG 1 Mapped Recommendations

Recommendation		Set Correctly	
		Yes	No
1.1	Ensure a separate user and group exist for Cassandra	☐	☐
1.5	Ensure the Cassandra service is run as a non-root user	☐	☐
3.1	Ensure the cassandra and superuser roles are separate	☐	☐
3.2	Ensure that the default password is changed for the cassandra role	☐	☐
3.4	Ensure that Cassandra is run using a non-privileged, dedicated service account	☐	☐
3.5	Ensure that Cassandra only listens for network connections on authorized interfaces	☐	☐
3.8	Review Superuser/Admin Roles	☐	☐
4.2	Ensure that auditing is enabled	☐	☐

Appendix: CIS Controls v8 IG 2 Mapped Recommendations

Recommendation		Set Correctly	
		Yes	No
1.1	Ensure a separate user and group exist for Cassandra	☐	☐
1.2	Ensure the latest version of Java is installed	☐	☐
1.3	Ensure the latest version of Python is installed	☐	☐
1.4	Ensure latest version of Cassandra is installed	☐	☐
1.5	Ensure the Cassandra service is run as a non-root user	☐	☐
1.6	Ensure clocks are synchronized on all nodes	☐	☐
2.1	Ensure that authentication is enabled for Cassandra databases	☐	☐
2.2	Ensure that authorization is enabled for Cassandra databases	☐	☐
3.1	Ensure the cassandra and superuser roles are separate	☐	☐
3.2	Ensure that the default password is changed for the cassandra role	☐	☐
3.4	Ensure that Cassandra is run using a non-privileged, dedicated service account	☐	☐
3.5	Ensure that Cassandra only listens for network connections on authorized interfaces	☐	☐
3.6	Ensure that Data Center Authorizations is activated	☐	☐
3.8	Review Superuser/Admin Roles	☐	☐
4.1	Ensure that logging is enabled.	☐	☐
4.2	Ensure that auditing is enabled	☐	☐
5.1	Inter-node Encryption	☐	☐
5.2	Client Encryption	☐	☐

Appendix: CIS Controls v8 IG 3 Mapped Recommendations

Recommendation		Set Correctly	
		Yes	No
1.1	Ensure a separate user and group exist for Cassandra	☐	☐
1.2	Ensure the latest version of Java is installed	☐	☐
1.3	Ensure the latest version of Python is installed	☐	☐
1.4	Ensure latest version of Cassandra is installed	☐	☐
1.5	Ensure the Cassandra service is run as a non-root user	☐	☐
1.6	Ensure clocks are synchronized on all nodes	☐	☐
2.1	Ensure that authentication is enabled for Cassandra databases	☐	☐
2.2	Ensure that authorization is enabled for Cassandra databases	☐	☐
3.1	Ensure the cassandra and superuser roles are separate	☐	☐
3.2	Ensure that the default password is changed for the cassandra role	☐	☐
3.3	Ensure there are no unnecessary roles or excessive privileges	☐	☐
3.4	Ensure that Cassandra is run using a non-privileged, dedicated service account	☐	☐
3.5	Ensure that Cassandra only listens for network connections on authorized interfaces	☐	☐
3.6	Ensure that Data Center Authorizations is activated	☐	☐
3.7	Review User-Defined Roles	☐	☐
3.8	Review Superuser/Admin Roles	☐	☐
4.1	Ensure that logging is enabled.	☐	☐
4.2	Ensure that auditing is enabled	☐	☐
5.1	Inter-node Encryption	☐	☐
5.2	Client Encryption	☐	☐

Appendix: CIS Controls v8 Unmapped Recommendations

Recommendation		Set Correctly	
		Yes	No
	No unmapped recommendations to CIS Controls v8.0	☐	☐

Appendix: Change History

Date	Version	Changes for this version